Hanski, who argued that a population of a species is made up of differing, dynamic components. One part of a population may become extinct, while another thrives. The thriving element may subsequently help reestablish the population that has died out.

In the process, British ecologist John Odling-Smee argued, so-called “niche-constructor” species create a more favorable environment for themselves—as seen in countless examples, from ancient oxygen-producing cyanobacteria that altered the composition of the atmosphere in prehistoric times, to beavers creating wetlands.

Modern methods

Traditionally, the task of monitoring environmental change has been the responsibility of academics and professional ecologists, but millions of interested amateurs now provide enormous amounts of raw data on everything from flowering dates to butterfly numbers, and from the state of coral reefs to the breeding populations of birds. With computer power to quickly process vast amounts of data, and with Earth’s ecology changing faster than ever, this “citizen science” looks set to become an invaluable resource for ecology.